



Introduction to Database

Objectives

- Understand the importance of SQL for querying data in databases.
- Present the steps on how to install MySQL.
- Discuss basic SQL Syntax
- Perform simple commands such as creating databases, tables, and querying data from tables.
- Perform an activity pertaining to the topics discussed.

What is SQL?

- SQL is a coding language for querying data in databases.

Databases vs. Excel

1

Databases can process more data than spreadsheets. Whereas Excel can handle up to 1 million rows of data, a database like MySQL can handle over a billion rows.

2

Databases can interact with other programming languages such as Python, Ruby, or Java, allowing you to execute many more powerful tasks and queries with your data.

3

Databases can connect to the internet, allowing millions of users to access them at the same time.

Why learn SQL?

SQL is used by Google, Amazon, Apple, Airbnb, Netflix, Uber, Wells Fargo, Shopify, WordPress, and hundreds of thousands of other companies. In short, SQL is everywhere!

SQL job skills are in high demand

SQL is one of the most robust and easy to use programming languages.

Among the most in-demand jobs where learning SQL is a must...

Data Scientist

Data Analyst

Software Engineer

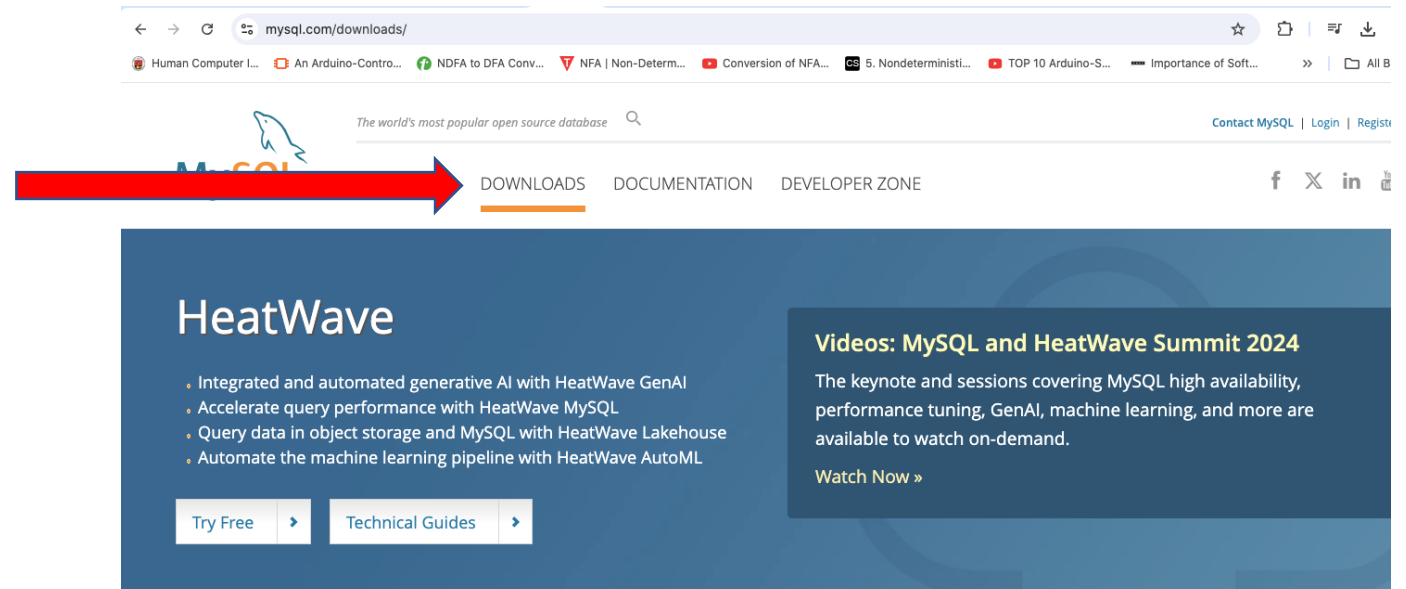
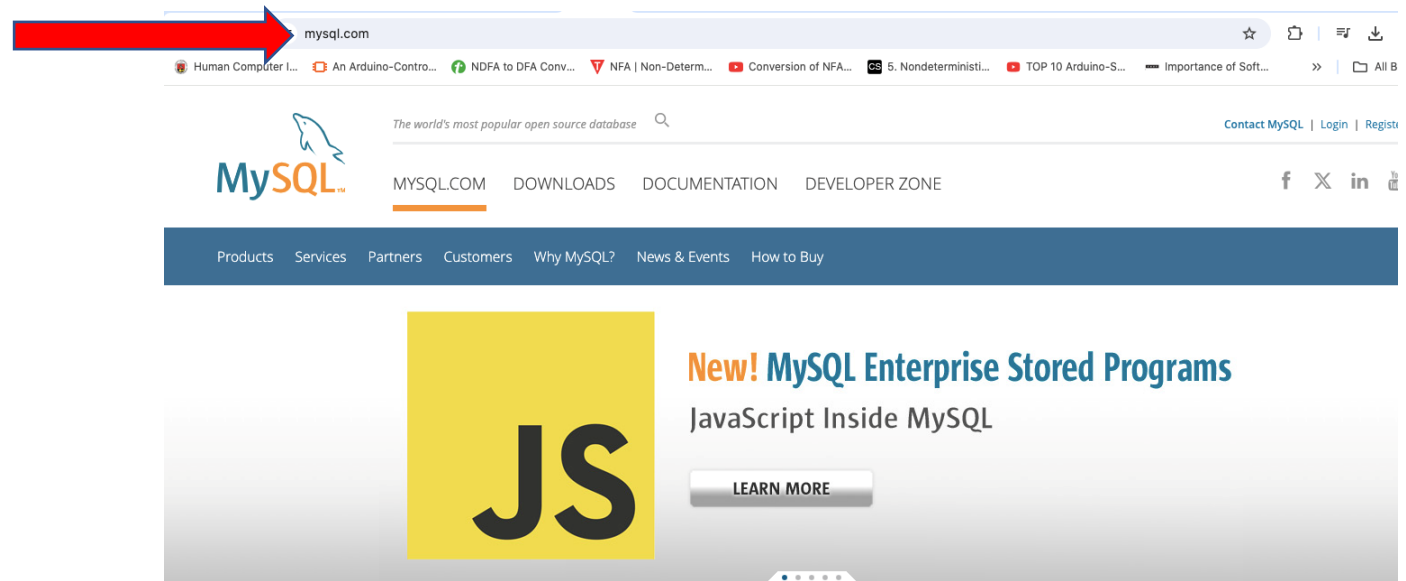


Installing MySQL



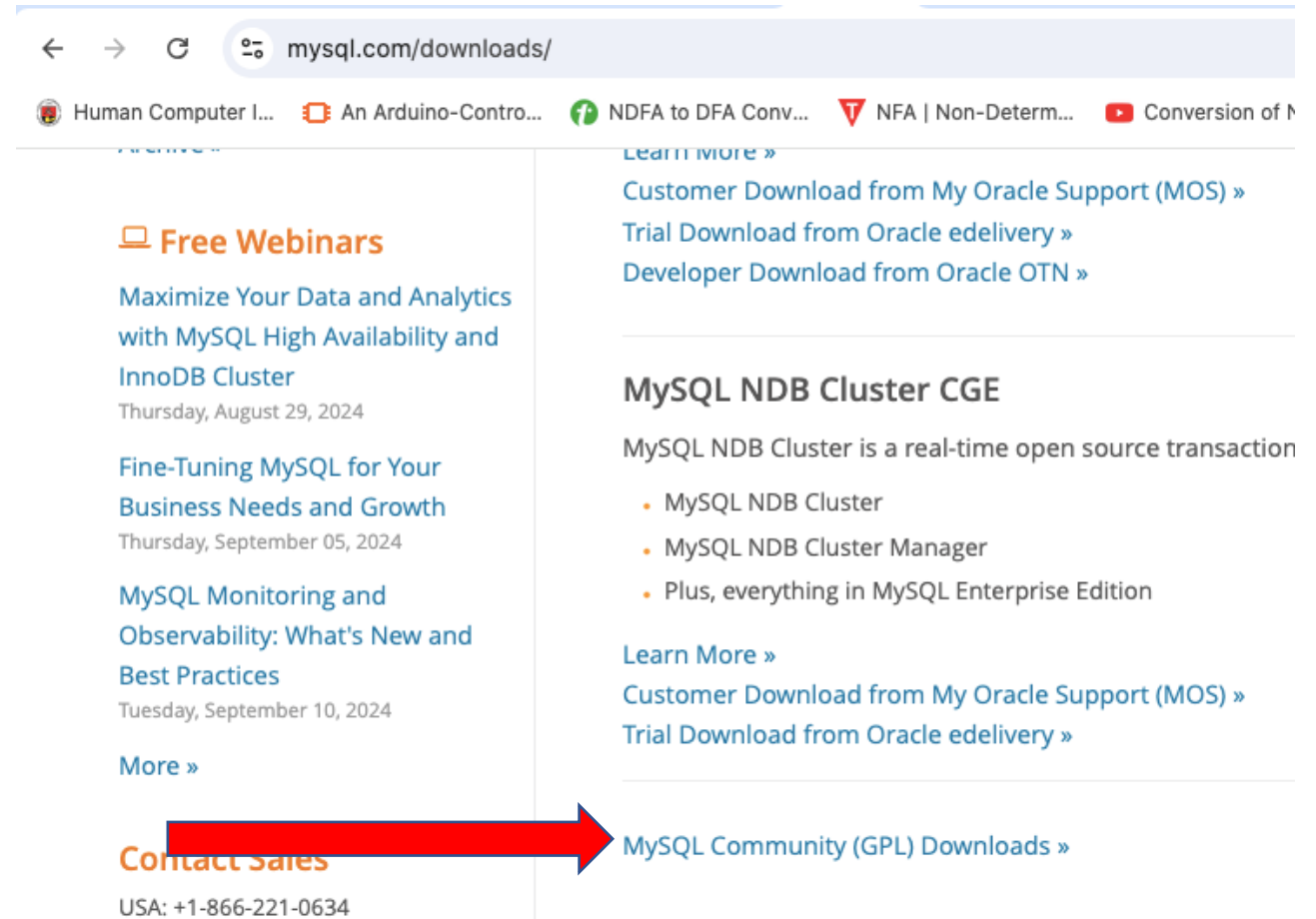
Installation

1. Go to mysql.com website
2. Click Downloads



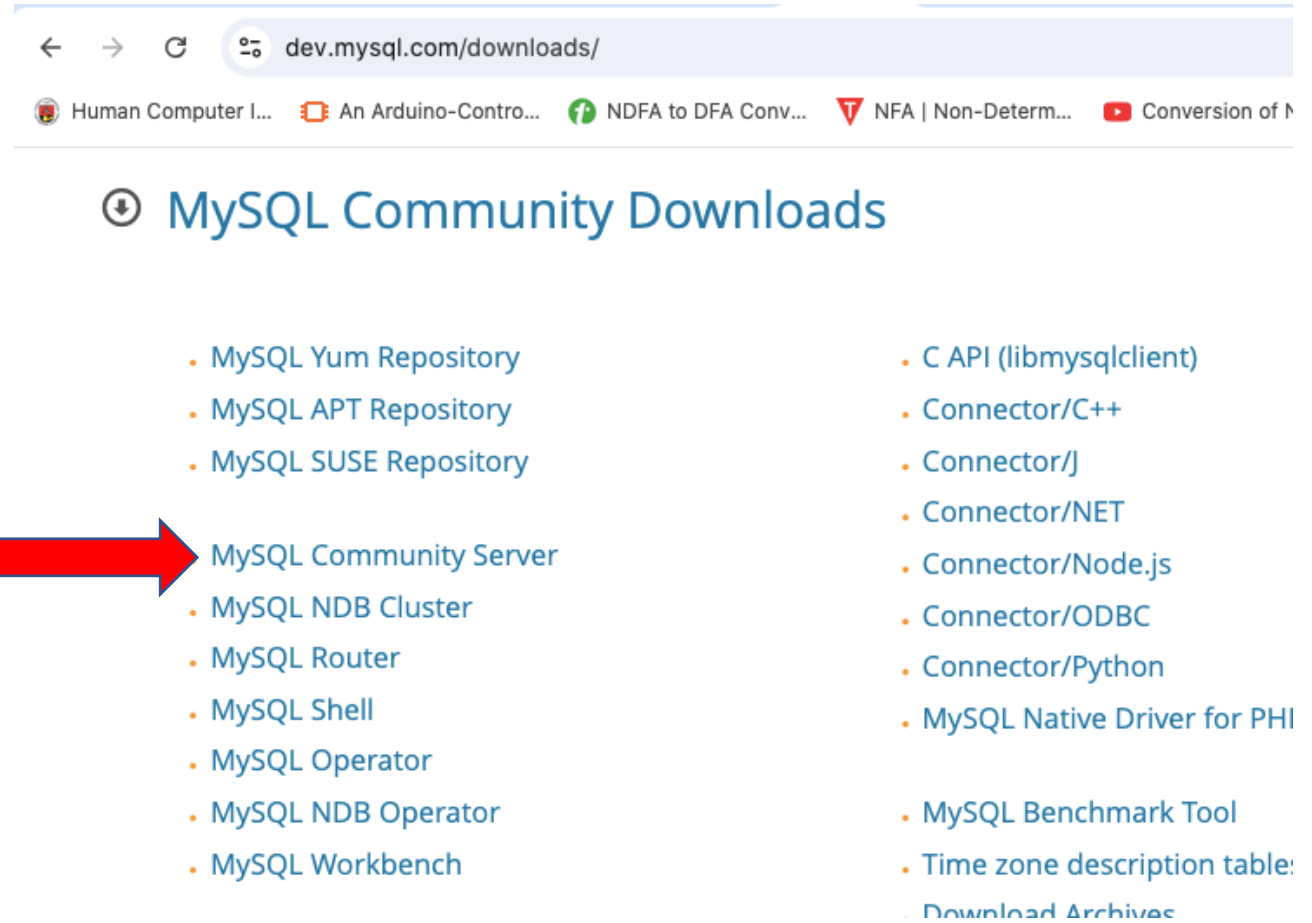
Downloading the Installer

3. From the Downloads page, scroll down and click MySQL Community Downloads



Downloading the Installer

4. Click Community Server



The screenshot shows a web browser at the URL `dev.mysql.com/downloads/`. The page title is "MySQL Community Downloads". A large red arrow points to the "MySQL Community Server" link in the list of download options.

- MySQL Yum Repository
- MySQL APT Repository
- MySQL SUSE Repository
- MySQL Community Server
- MySQL NDB Cluster
- MySQL Router
- MySQL Shell
- MySQL Operator
- MySQL NDB Operator
- MySQL Workbench
- C API (libmysqlclient)
- Connector/C++
- Connector/J
- Connector/NET
- Connector/Node.js
- Connector/ODBC
- Connector/Python
- MySQL Native Driver for PHP
- MySQL Benchmark Tool
- Time zone description table
- Download Archives

Downloading the Installer

5. The website automatically detects your operating system, but you can select the version, the operating system, and the OS version.

6. At the bottom, scroll down to select the version appropriate to your OS.

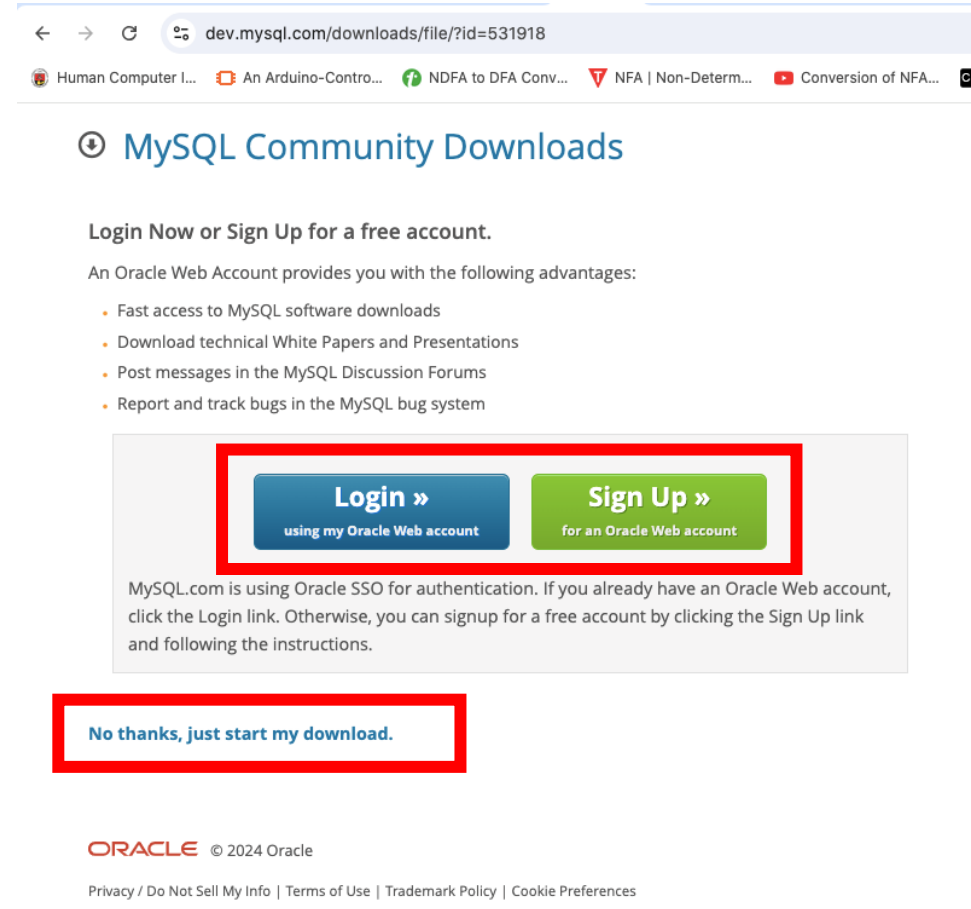
7. Click Download.

The screenshot shows the MySQL Community Downloads page. The browser address bar displays `dev.mysql.com/downloads/mysql/`. The page title is "MySQL Community Downloads" with a sub-header "MySQL Community Server". There are two tabs: "General Availability (GA) Releases" (selected) and "Archives". A red box highlights the selection area with three dropdown menus: "Select Version:" set to "9.0.1 Innovation", "Select Operating System:" set to "macOS", and "Select OS Version:" set to "All". Below this, another red box highlights a table of download links. A warning message states: "Packages for Sonoma (14) are compatible with Ventura (13)".

Operating System	Version	Size	Action
macOS 14 (ARM, 64-bit), DMG Archive	9.0.1	529.0M	Download
(mysql-9.0.1-macos14-arm64.dmg) MD5: 1379aaa58aa414aad00016713cf574bb Signature			
macOS 14 (x86, 64-bit), DMG Archive	9.0.1	533.7M	Download

Downloading the Installer

8. You can Sign-up or Log-in. But if you don't want to, click "No thanks, just start my download".



The screenshot shows a web browser window with the URL `dev.mysql.com/downloads/file/?id=531918`. The page title is "MySQL Community Downloads". Below the title, there is a section for "Login Now or Sign Up for a free account." which lists advantages of an Oracle Web Account. Two buttons, "Login »" and "Sign Up »", are highlighted with a red box. Below these buttons, a paragraph explains the use of Oracle SSO for authentication. At the bottom, a link "No thanks, just start my download." is also highlighted with a red box. The footer contains the Oracle logo, copyright information for 2024, and links to privacy and terms of use.

dev.mysql.com/downloads/file/?id=531918

Human Computer I... An Arduino-Contro... NDFA to DFA Conv... NFA | Non-Determ... Conversion of NFA...

MySQL Community Downloads

Login Now or Sign Up for a free account.

An Oracle Web Account provides you with the following advantages:

- Fast access to MySQL software downloads
- Download technical White Papers and Presentations
- Post messages in the MySQL Discussion Forums
- Report and track bugs in the MySQL bug system

Login »
using my Oracle Web account

Sign Up »
for an Oracle Web account

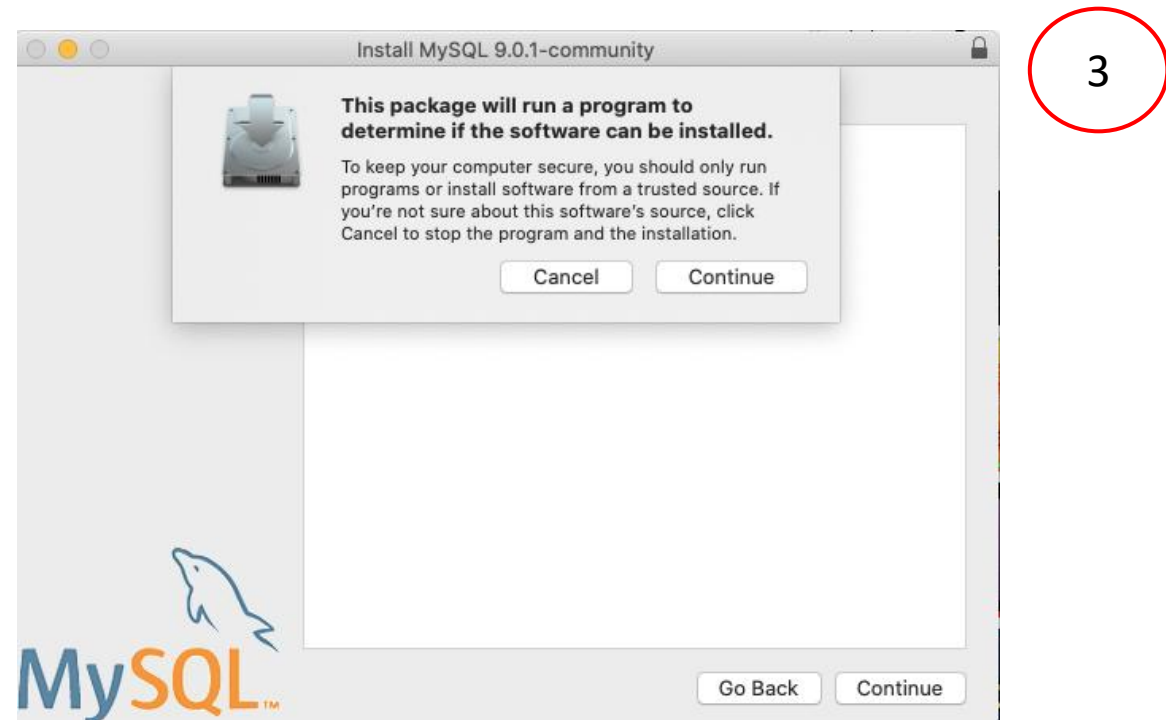
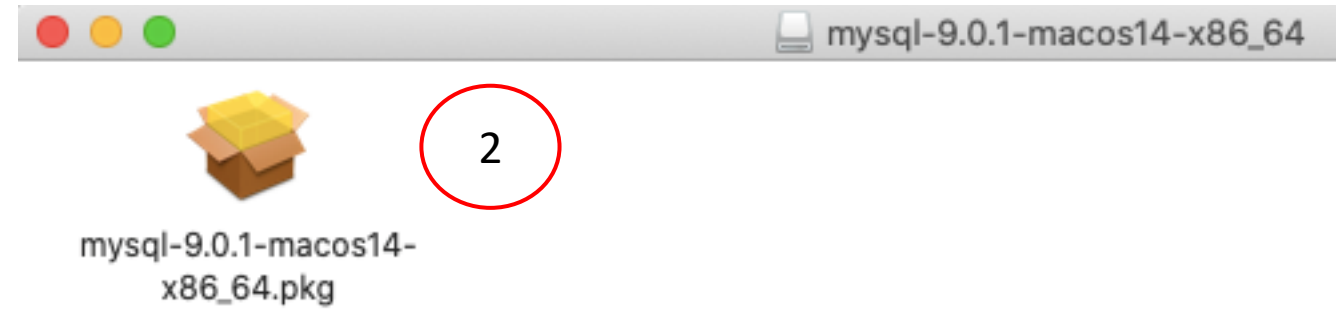
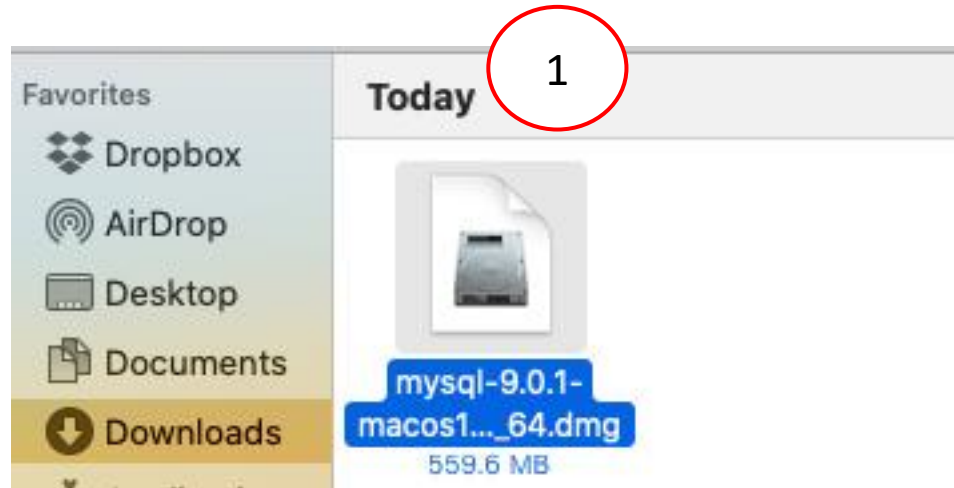
MySQL.com is using Oracle SSO for authentication. If you already have an Oracle Web account, click the Login link. Otherwise, you can signup for a free account by clicking the Sign Up link and following the instructions.

[No thanks, just start my download.](#)

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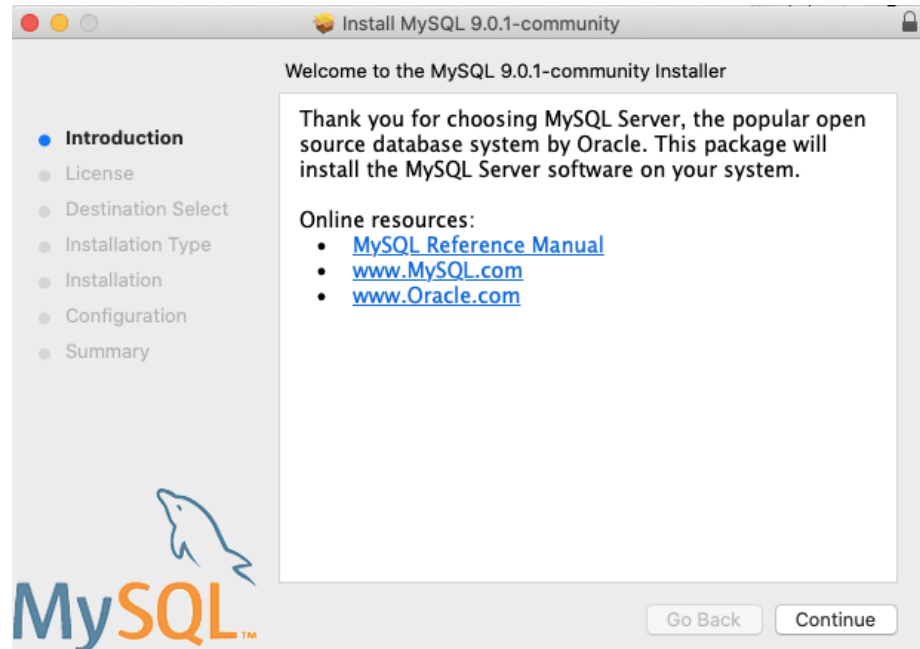
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Installation

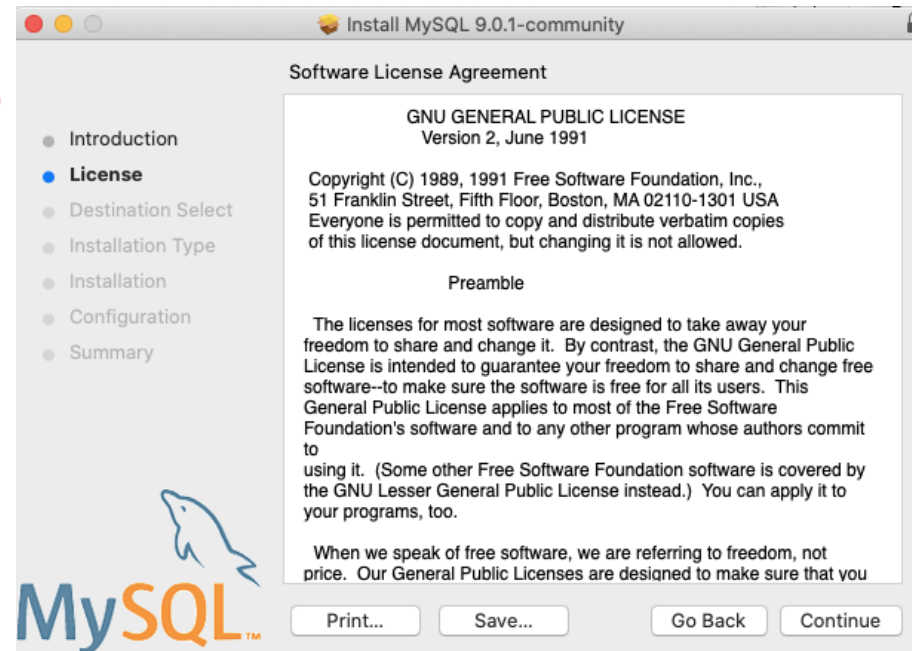


Installation

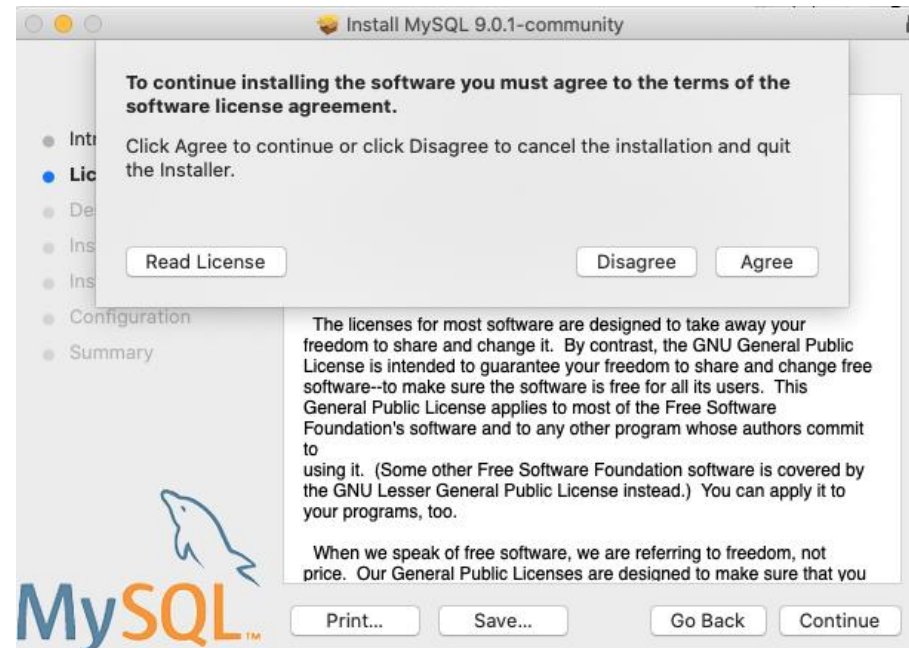
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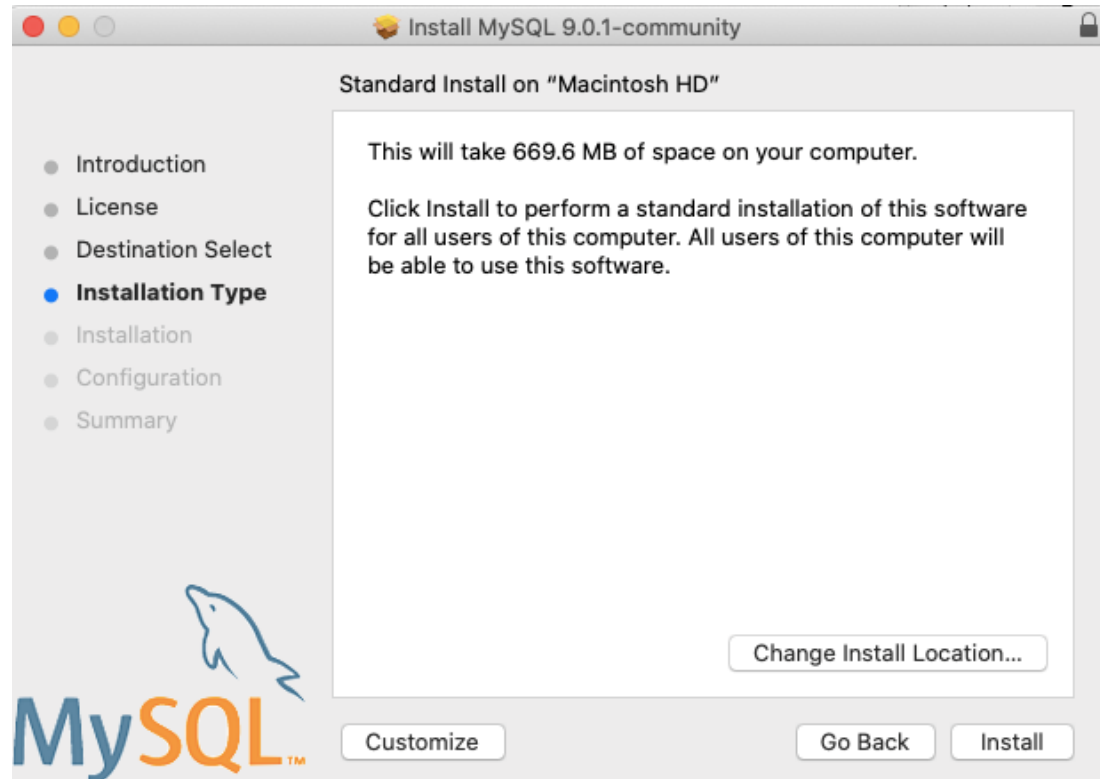


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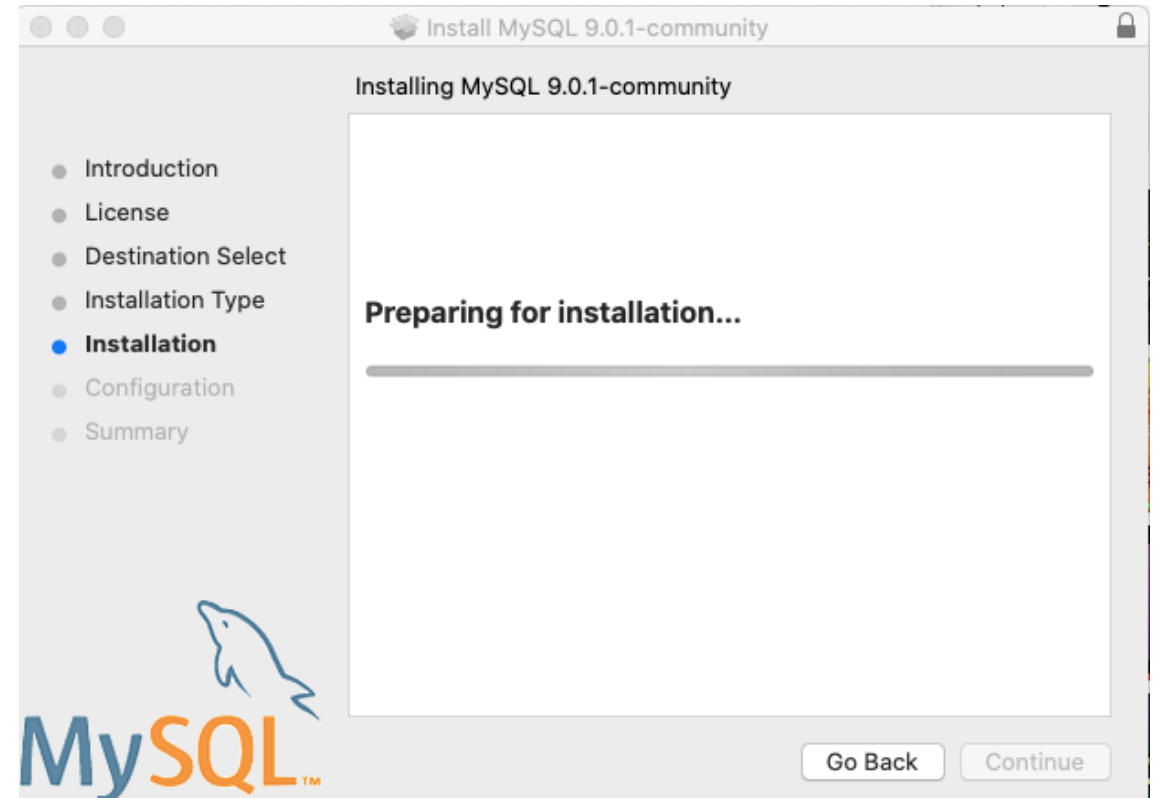


Installation

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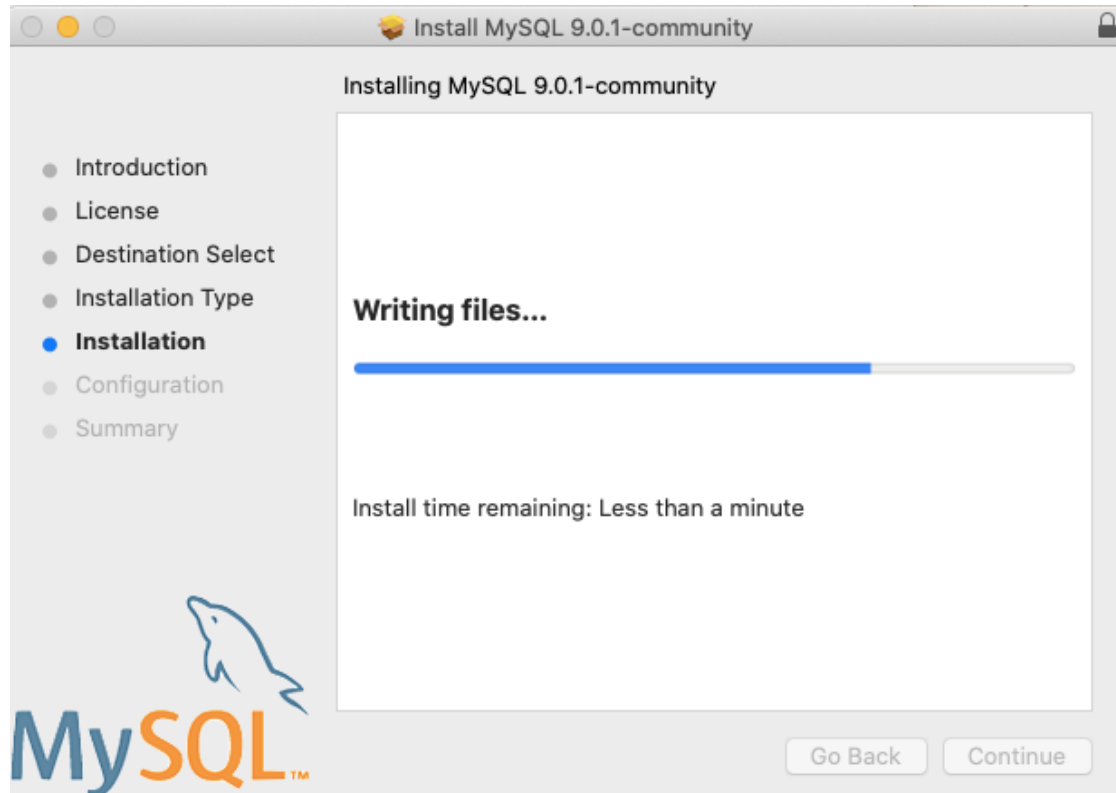


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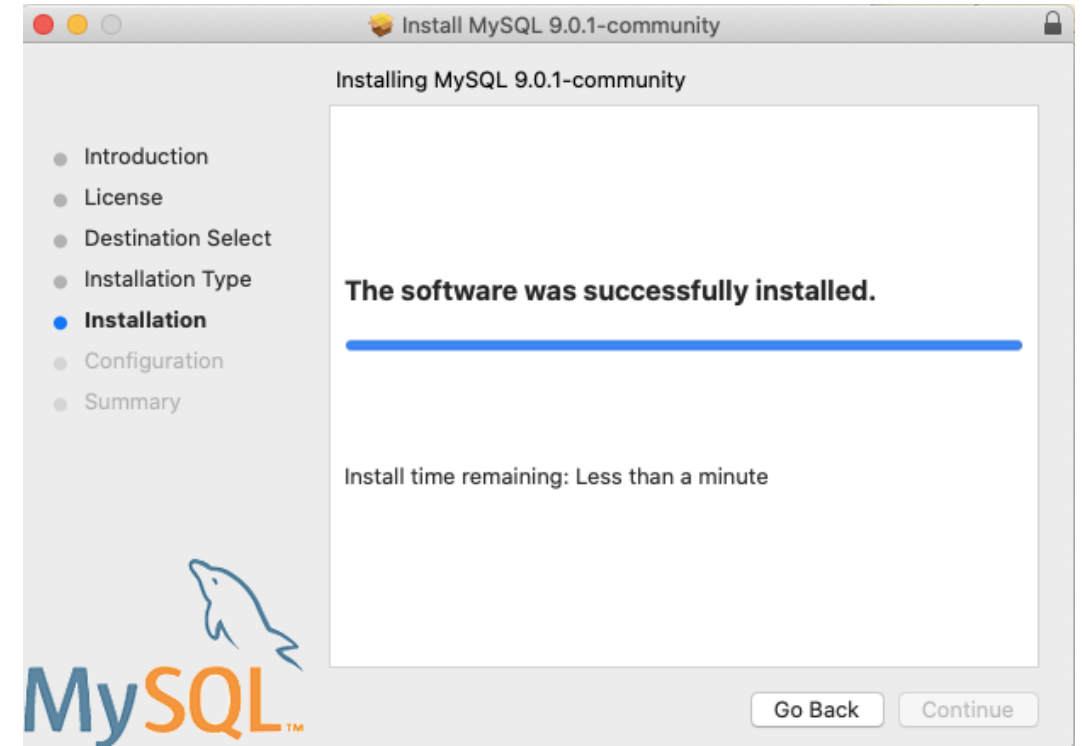


Installation

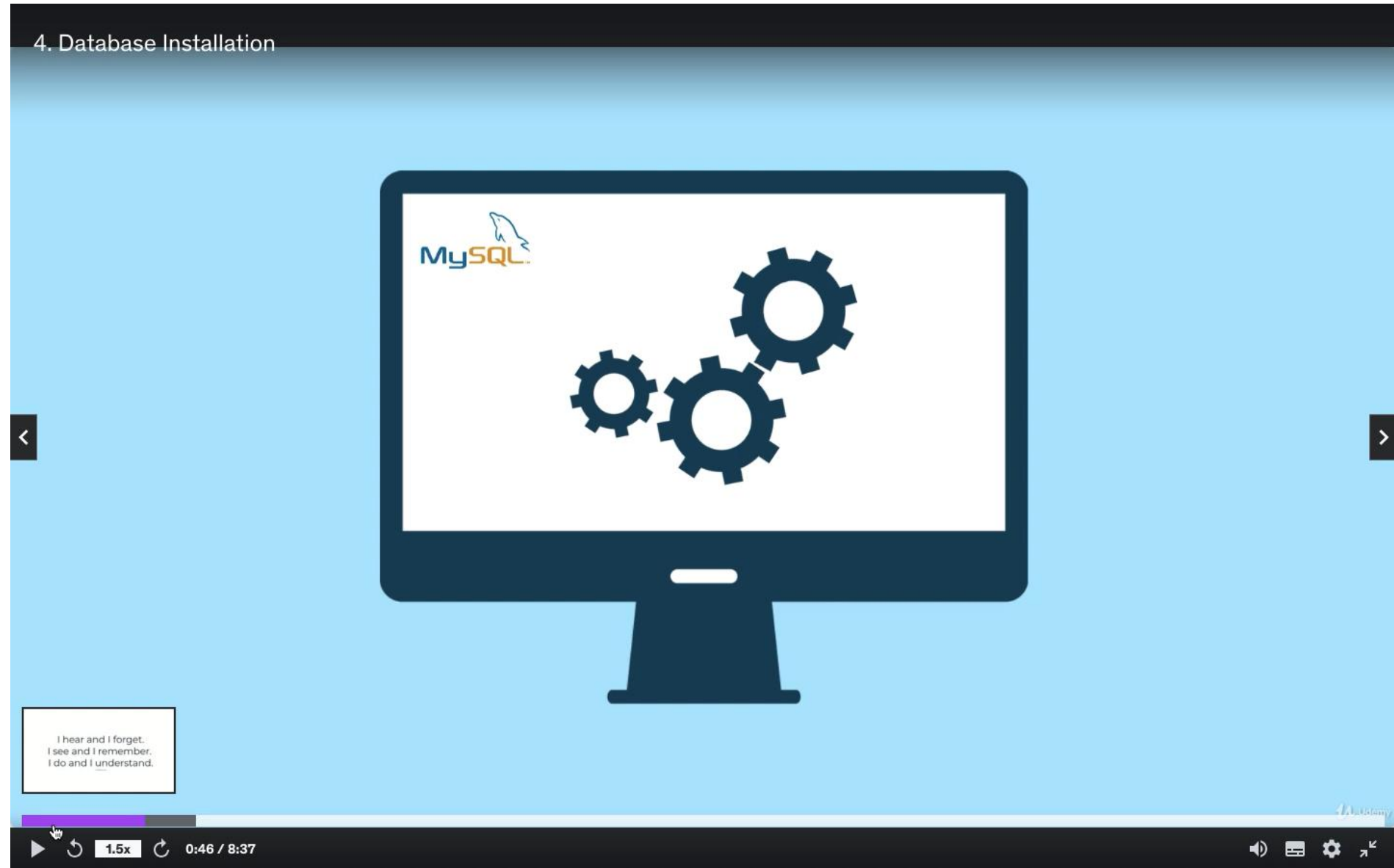
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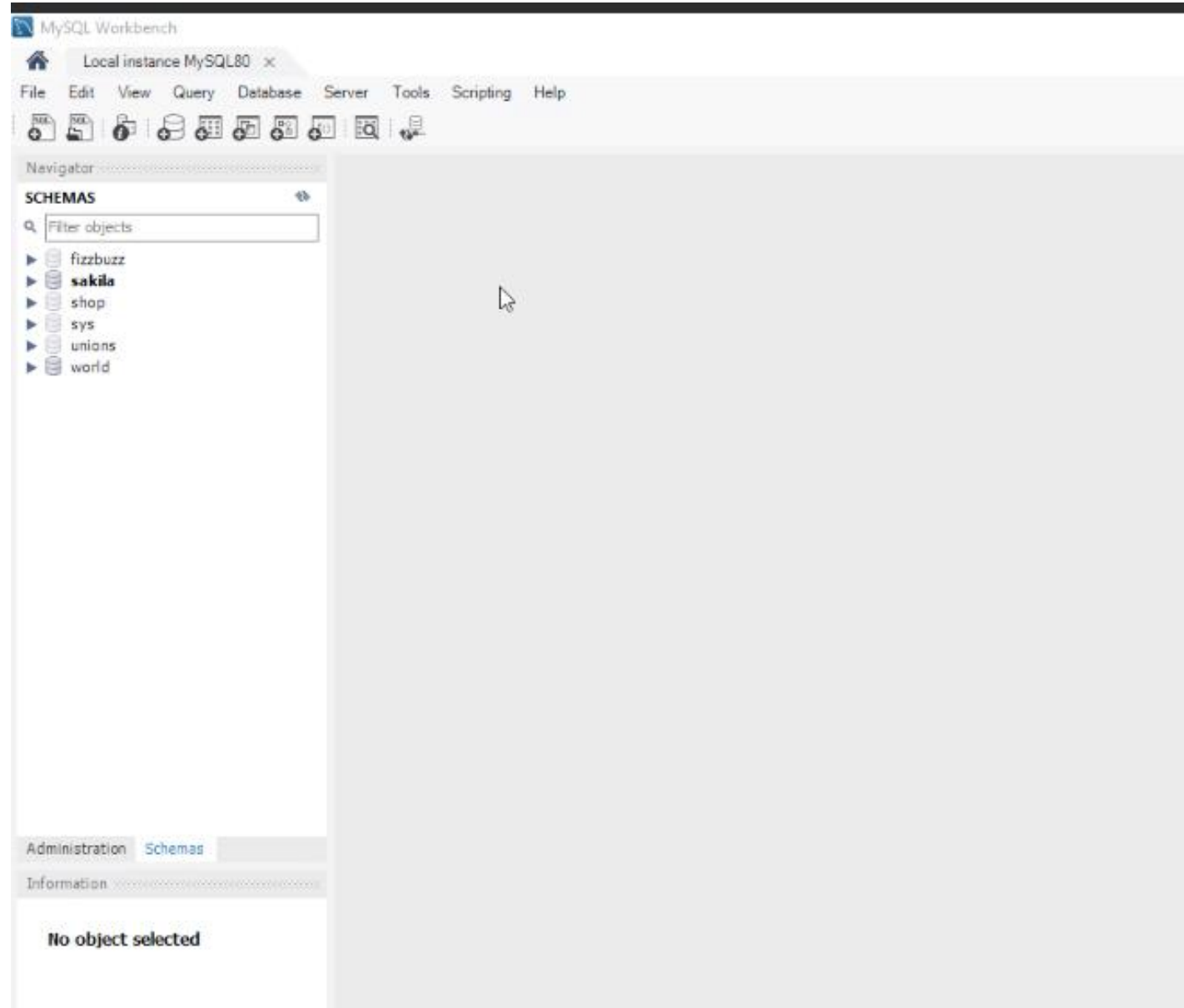
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MySQL Windows Installation



MySQL Windows Installation





Database Fundamentals



Important points to remember

- Relational databases contain organized data.
- In MySQL, a database can also be referred to as a Schema.
- Schemas contain objects such as Tables, Columns, Properties, Views, Data types, Indexes & Stored Procedures.
- An Entity-Relationship Diagram can be used to visualize the organization of a database.

Elements



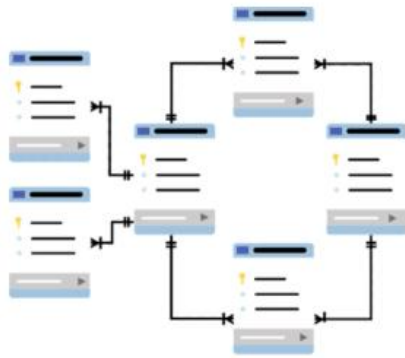
Schema

- In MySQL, the word schema and database are used interchangeably.

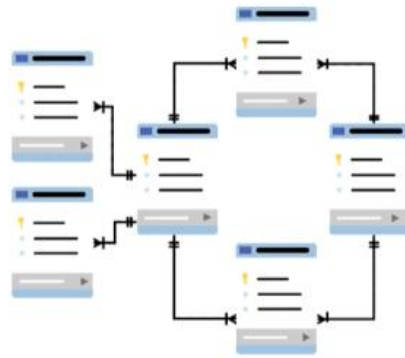


SCHEMA = DATABASE

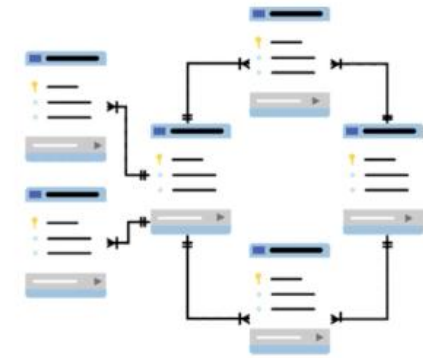
What is Schema?



SCHEMA 1



SCHEMA 2

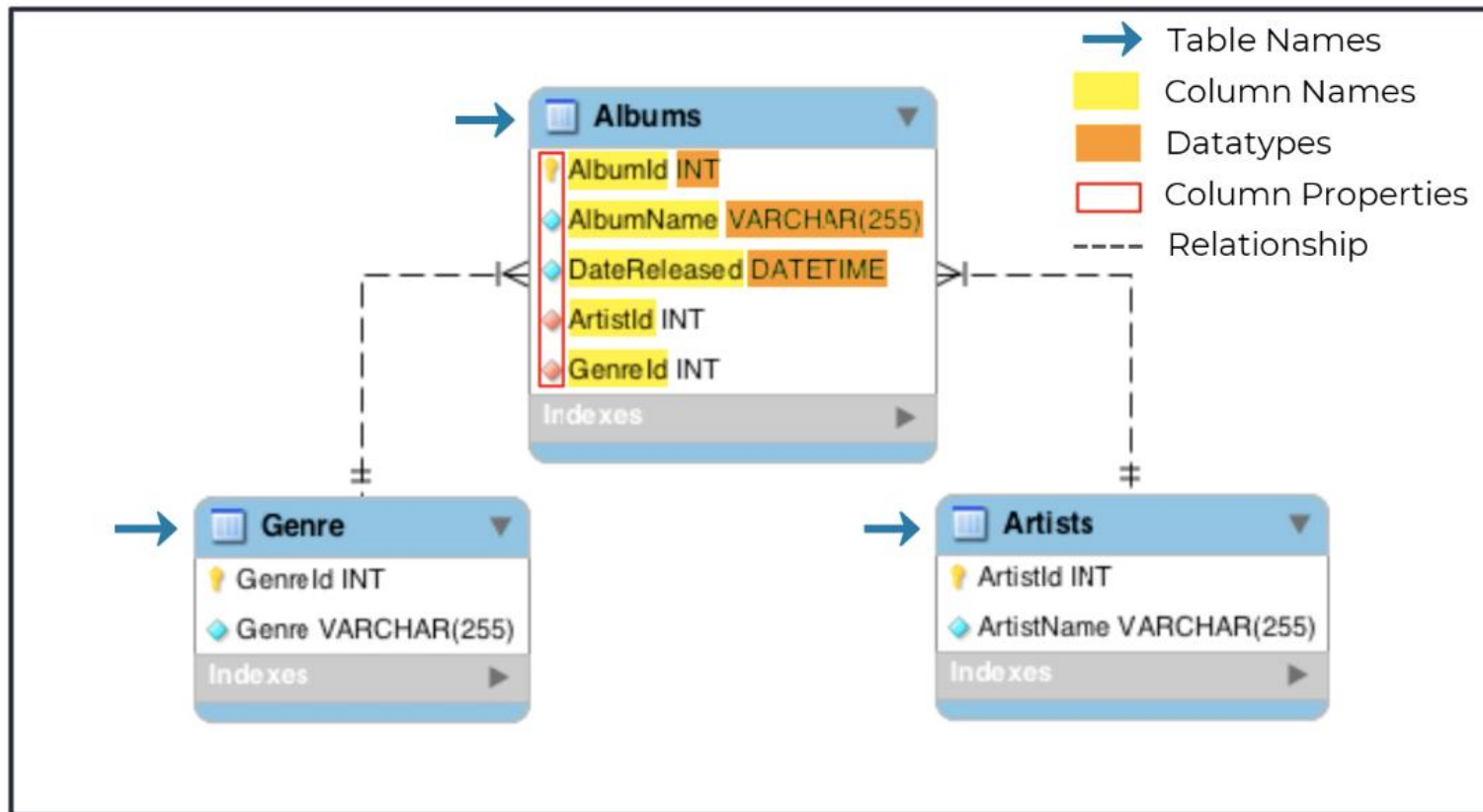


SCHEMA 3

"Organisation of
data in a database"

An example of a schema (in general)

MUSIC



Difference between Schema and Database

- A database (often called DB) is an organized structure designed to store, modify, and process related information, mostly in large volumes. Databases are actively used for dynamic sites with significant amounts of data.
- On the contrary, the term database schema can mean either a visual data representation, a set of rules it is subject to, or a complete set of objects owned by a particular user.

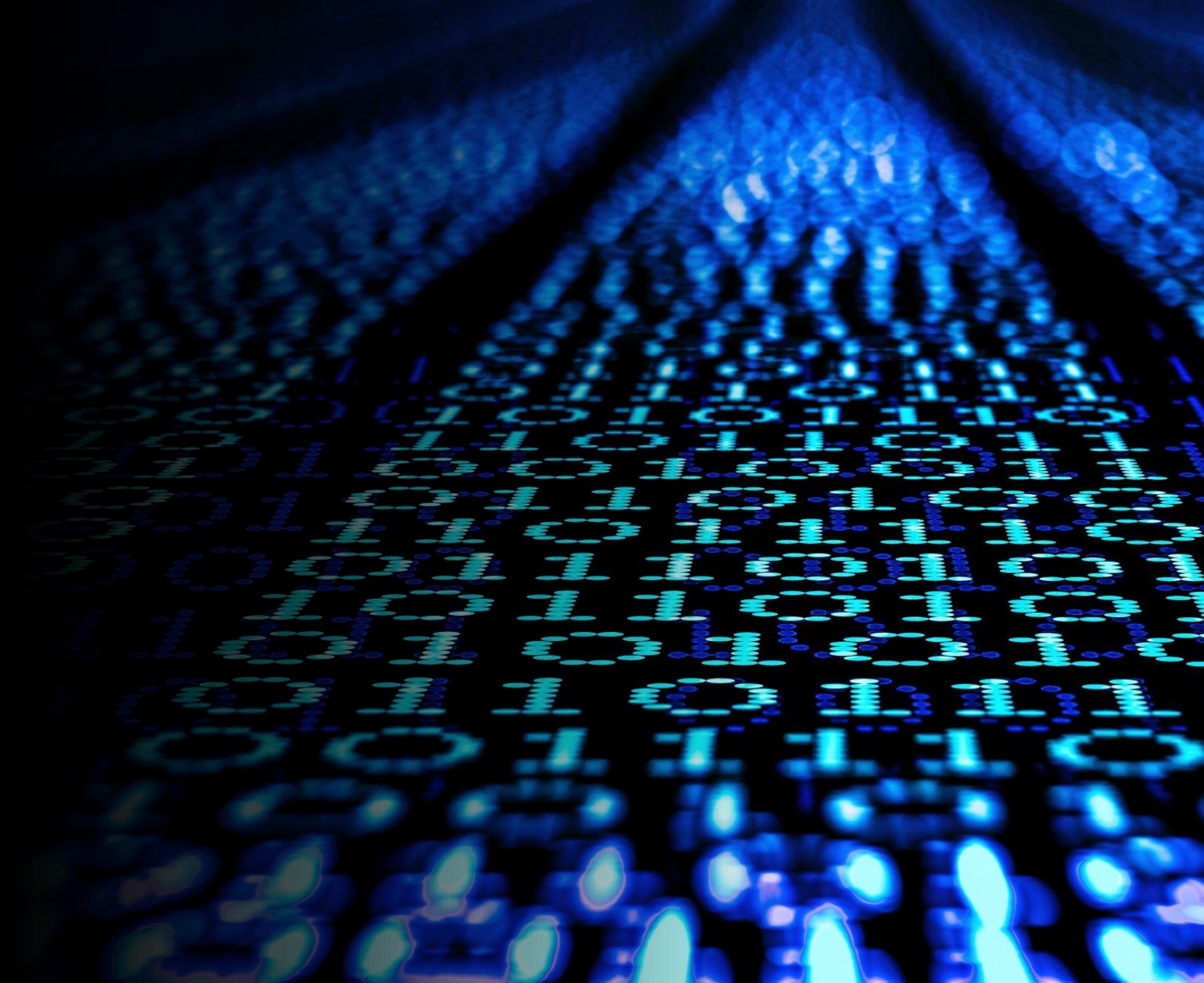
Difference between Schema and Database

- A database is more about data and content, while a schema is more about the structure of the said data.

COMPARISON BASIS	DATABASE	SCHEMA
DEFINITION	The database is an application that stores the organized collection of interrelated data	The schema is a logical representation of a database
STATEMENT	A DML is used to generate and modify the records in the database	A DDL statement is used to define the schema for a database structure
INCLUDE	A database is an organized collection of structured information or data	A schema describes how the metadata is set up, and usually addresses standards for common components of metadata like dates, names, and places
MEMORY	It uses memory to store data	It does not use memory to store data



SQL Syntax



Is SQL Case Sensitive?

- SQL keywords are NOT case sensitive: select is the same as SELECT

Single line and multi-line comments

- SQLyog, as a MySQL and MariaDB client, supports the standard SQL comment syntaxes.

There are two primary ways to add comments in SQLyog:

- ***Single-Line Comments:***
- Begin the comment with two hyphens followed by a space (`--`).
- The comment extends to the end of the line.

Single line and multi-line comments

Single-Line Comments:

- Example

```
SELECT * FROM users; -- This retrieves all users
```

Multi-line comments

- Multi-Line (Block) Comments: Begin the comment with a forward slash and an asterisk (/ *).
- The comment can span multiple lines.
- End the comment with an asterisk and a forward slash (* /).

Multi-line comments

- Example:

```
/*  
  This is a multi-line comment  
  explaining the following query.  
*/  
SELECT name, email FROM customers WHERE status = 'active';
```